

We take a look at the three most common wireless standards available today—infra-red, Bluetooth and WiFi. Learn how they work, and how to use them

Beyond Wires

The need for wireless communications arose when people started using more than one electronic device to store personal data, such as contacts, photos, appointments, etc. All this data was stored all over the place: in laptops, PDAs, mobile phones, and even in wrist watches. Using wires for communication beats the very purpose of mobile devices. So, the first wireless standard to gain acceptance in the market was infra-red. Then Bluetooth arrived, and it is slowly but steadily entering the market. These standards, however, had low data transfer rates; so a new standard named WiFi was born. It supports high data rates and is used primarily in laptops for network access.

Infra-red

Infra-red is similar to normal light and is just another electromagnetic wave. It is called infra-red because the frequency and

wavelength takes it below the red end of the spectrum, making it invisible to humans. Infra-red is the simplest and one of the most prevailing wireless communication standards. It uses an infra-red beam to transfer data between devices. Remote controls for various devices—televisions, CD players, air conditioners, etc.—all use this technology. IrDA (Infra-red Data Association) is an organisation that maintains the standard for communication via Infra-red, and there are currently more than 100 member companies and many more devices that have an IrDA port.

Like any other standard, IrDA is a cable replacement standard; it creates a virtual wire using an infra-red beam to transfer data. It is a half-duplex, point to point protocol—only one device can transmit data at a time. Also, the devices need to have a clear line of sight between their ports to function—the IrDA ports of

the communicating devices must face each other without obstructions. The two devices that take part in this communication are called primary and secondary devices. Primary devices are the ones that



Sony Ericsson T310 is a popular IrDA enabled mobile phone. The black rectangle seen here is the infra-red port